(i) Let $d = 1 - p_A - p_B$
The match is won by A after r draws followed by an A win.

$$\begin{aligned} P(A \text{ wins}) &= p_A + dp_A + d^2 p_A + \cdots \\ &= p_A(1 + d + d^2 + \cdots) \\ &= \frac{p_A}{1-d} \\ &= \frac{p_A}{p_A + p_B} \end{aligned}$$

(ii) After each game, the difference between the number of games won by A and B changes by 1. The match ends when this difference is 2 or -2.
First two games:

$$AA: \ p^2 \Rightarrow A \text{ wins}$$

$$BB: \ q^2 \Rightarrow B \text{ wins}$$

$$AB \text{ or } BA: \ 2pq \Rightarrow \text{back to } 0$$

$$\begin{aligned} P_A &= p^2 + 2pqP_A \\ &= \frac{p^2}{1-2pq} \\ &= \frac{p^2}{p^2+q^2} \end{aligned}$$

(iii)
Cautious version
Let C_i be the probability of winning with i tokens.

$$C_0 = 0, \ C_4 = 1$$

$$C_i = pC_{i+1} + qC_{i-1} \quad (1 \le i \le 3)$$

$$C_1 = pC_2, \quad C_2 = pC_3 + qC_1, \quad C_3 = p + qC_2$$

$$\begin{aligned} C_2 &= p(p + qC_2) + qC_1 \\ &= p^2 + pqC_2 + qC_1 \\ &= p^2 + pqC_2 + pqC_2 \\ &= \frac{p^2}{p^2 + q^2} \end{aligned}$$

$$C_1 = pC_2 = \frac{p^3}{p^2 + q^2}$$

Bold version

The player must double twice before losing all tokens.

$$B = p^2$$

Comparison,

$$\begin{aligned} C_1 - B &= \frac{p^3}{p^2 + q^2} - p^2 \\ &= \frac{p^2(p - p^2 - q^2)}{p^2 + q^2} \\ &= \frac{p^2 q(p - q)}{p^2 + q^2} \end{aligned}$$

$$p - q > 0 \iff p > \frac{1}{2}$$

So cautious is more likely if $\frac{1}{2} < p < 1$

bold is more likely if $0 < p < \frac{1}{2}$

This solution was submitted by **Jon**

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